Analysis Notes

Kota Uezono

June 8, 2026

Theorem 0.1 (Hölder's inequality)**.** *Let* $p, q \in [1, \infty]$ *satisfy* $\frac{1}{p} + \frac{1}{q} = 1$ *and let* f *and* g *be measurable functions. Then*

$$\|fg\|_1 \leq \|f\|_p \|g\|_q$$

Proof. The case $p = \infty$ is trivial. Assume $p, q < \infty$. We first prove the following auxiliary inequality: If $s, t > 0$, then

$$st \leq \frac{s^p}{p} + \frac{t^q}{q}.$$

Let $\sigma, \tau \in \mathbb{R}$, and set $s = e^\sigma, t = e^\tau$. Since e^x is convex and

$$\frac{1}{p} + \frac{1}{q} = 1$$

we obtain

$$e^{\sigma+\tau} \leq \frac{e^{p\sigma}}{p} + \frac{e^{q\tau}}{q}.$$

Hence

$$st \leq \frac{s^p}{p} + \frac{t^q}{q}.$$

Using this inequality for $(s, t) = (|f|, |g|)$, we obtain

$$|fg| \leq \frac{|f|^p}{p} + \frac{|g|^q}{q}.$$

Integrating both sides, we obtain

$$\|fg\|_1 \leq \frac{\|f\|_p^p}{p} + \frac{\|g\|_q^q}{q}. \tag{1}$$

Setting $C = \|f\|_p \|g\|_q, \tilde{f} = \frac{f}{\|f\|_p}, \tilde{g} = \frac{g}{\|g\|_q}$, we obtain

$$\|\tilde{f}\|_p^p = \|\tilde{g}\|_q^q = 1$$

The case $C = \infty$ is trivial. So, let $C < \infty$. Then,

$$C^{-1}\|fg\|_1 = \|\tilde{f}\tilde{g}\|_1 \overset{(1)}{\leq} \frac{\|\tilde{f}\|_p^p}{p} + \frac{\|\tilde{g}\|_q^q}{q} = \frac{1}{p} + \frac{1}{q} = 1.$$

Multiplying both sides by C, we obtain

$$\|fg\|_1 \leq \|f\|_p \|g\|_q.$$

□

Theorem 0.2 (Minkowski's inequality)**.** *Let $p \in [1, \infty]$, and $f, g : S \to \mathbb{C}$ is measurable functions. Then,*

$$\|f + g\|_p \leq \|f\|_p + \|g\|_p$$

Proof. The case $p = \infty$ and the case $p = 1$ follow directly from the triangle inequality. Assume $1 < p < \infty$. We first prove the following auxiliary inequalities: Set $h = |f + g|$. Then,

$$\|h\|_p^p \leq 2^{p-1}(\|f\|_p^p + \|g\|_p^p). \tag{2}$$

$$\|h\|_p^p \leq (\|f\|_p + \|g\|_p)\|h\|_p^{p-1}. \tag{3}$$

Proof of (2): Consider the inequality:

$$s, t \in \mathbb{C}, |s + t|^p \leq 2^{p-1}(|s|^p + |t|^p)$$

By triangle inequality,

$$|s + t| \leq |s| + |t|$$

So, it suffices to prove the claim in the case $s, t \in [0, \infty)$. Since $x \mapsto x^p$ is convex, we obtain

$$\left(\frac{s + t}{2}\right)^p \leq \frac{s^p}{2} + \frac{t^p}{2}.$$

Multiplying both sides by 2^p, we obtain

$$(s + t)^p \leq 2^{p-1}(s^p + t^p)$$

Applying the above inequality pointwise to $s = f(x), t = g(x)$, and integrate both sides, we obtain

$$\|h\|_p^p \leq 2^{p-1}(\|f\|_p^p + \|g\|_p^p).$$

Proof of (3): Since

$$h = |f + g| \leq |f| + |g|,$$

we have

$$h^p = hh^{p-1} \leq (|f| + |g|)h^{p-1}.$$

Integrating both sides yields

$$\|h\|_p^p \leq \|fh^{p-1}\|_1 + \|gh^{p-1}\|_1.$$

Let $q = \frac{p}{p-1}$. Then,

$$\|h^{p-1}\|_q = \|h\|_p^{p-1}$$

By Hölder's inequality, we obtain

$$\|fh^{p-1}\|_1+\|gh^{p-1}\|_1 \overset{\text{Hölder}}{\leq} \|f\|_p\|h^{p-1}\|_q+\|g\|_p\|h^{p-1}\|_q = (\|f\|_p+\|g\|_p)\|h\|_p^{p-1}.$$

Having established (2) and (3), we now prove the desired inequality. The case $\|f\|_p + \|g\|_p = \infty$ is trivial. Assume $\|f\|_p + \|g\|_p < \infty$. Then $\|h\|_p < \infty$ by (2). If $\|h\|_p = 0$, the conclusion is immediate. Assume $\|h\|_p > 0$. Dividing both sides of (3) by $\|h\|_p^{p-1}$, we obtain

$$\|f + g\|_p \leq \|f\|_p + \|g\|_p.$$

□